Tone of Finnish politics in the news and on Twitter, 2013–2022

Technical report : Introduction to Data Science mini-project

Anniina, Rasmus, Takahiro: https://github.com/TakahiroOu/data_Science_Project · Blog: [https://medium.com/@anniinataiv/what-can-we-learn-from-finnish-politics-analysing-10-years-of-news-and-social-media-2aacd96b46c5]

1. Goal and target group

Our mini-project canvas defined the motivation to understand the gap between the news and political discussion on Twitter, for example signs of polarisation, and the target group as researchers and politicians. During the project, we extended the target group to also include the general audience interested in democracy as the results give interesting insights to them too. We asked: Does the tone of political texts differ between Yle's news and Finnish tweets, and how did it change between 2013 and 2022? We measured the proportion of positive, neutral and negative texts per year for ten political topics and nine parliamentary parties. The results for a general audience are in our blog, and this report describes the technical side and reflections for the project.

2. Data collection

Yle news. We used the Yle Finnish News Archive from Kielipankki (the Language Bank of Finland), distributed as VRT files: one token per line with its lemma and morphological analysis, and structural tags with metadata for each article, paragraph and sentence. We used the years 2013–2022 to match the Twitter data.

Tweets. Finnish tweets come from the Nordic Tweet Stream (Laitinen et al., 2018). We downloaded one CSV file per search term, with the tweet text and date. In total we used 61,750 tweets about the ten topics and 45,309 tweets about the parties.

3. Preprocessing

News leads (`preprocess_sentiment.py`, `preprocess_sentiment_2.py`). We parsed the VRT files line by line: the `<text>` tag gives the article metadata, and sentences with `paragraph_type="lead"` give the lead paragraph. We used leads because full articles would have made the ten-year dataset very large, and headlines were shorter, less informative and sometimes repeated as subheadings with the same tags. This would have created duplicates. For every lead we kept two versions: the **surface words** (VRT column 1) joined back into a sentence for the sentiment model, and the **lemmas** (VRT column 3) for keyword matching and TF-IDF. Because the corpus already contains lemmas, we did not need the stemming step planned in our canvas.

Our first version selected only articles with the metadata `main_department="Politiikka"`. This gave just **3,831 leads** for ten years (202–634 per year), because the department tag is missing for most articles before 2022. We therefore changed the selection to **politics-related lemmas**: a lead is kept if one of its lemmas starts with a political keyword or party name (e.g. *hallitus*, *presidentti*, *kokoomu*, *perussuomalai*). This gave **68,814 leads** for 2013–2022. Each lead is classified once, and the output is one JSON file per year with the lemmatized leads grouped by sentiment. There are some tricky words specially in parties such as the centre party, which give results which are not related to politics, but we tried to mitigate this by analysing the results and filtering them further if needed.

Tweets (`twitter_sentiment.py`). The Nordic Tweet Stream lets the user choose which fields to export, so we downloaded only the tweet text and date for each search term. This meant very little preprocessing was needed: we only converted HTML entities back to characters (e.g. `&` → `&`), kept tweets dated 2013–2022.

Cleaning for analysis. Lowercasing, removing URLs and characters other than Finnish letters, and normalising whitespace. For TF-IDF we also removed the NLTK Finnish stopwords and tokens shorter than three characters.

4. Methods

Sentiment classification. We used the pretrained Finnish BERT model `fergusq/finbert-finnsentiment` (Hugging Face Transformers, PyTorch) without fine-tuning. The model gives probabilities for positive, neutral and negative. We used a stricter rule than the most probable class: a text is neutral only if P(neutral) ≥ 0.98, otherwise it gets the more probable of positive and negative. We chose 0.98 because with lower thresholds most leads were classified as neutral. Our first version used only positive and negative. However, to compare polarisation in the news and on Twitter, we added the neutral class. News and tweets are classified with the same code (`sentiment_model.py`), so both follow exactly the same rule.

TF-IDF and keyword selection (`tfidf.py`). To find the most meaningful words, we computed TF-IDF over the first dataset of 3,831 politics leads (scikit-learn `TfidfVectorizer` and `CountVectorizer`, words in at least 2 and at most 50% of the leads) and compared it with plain word frequencies. TF-IDF ranks words that are frequent in a lead but not in all leads, so it gives better keyword candidates. From the top 50 words we left out words that are not political topics, such as first names and reporting verbs (*kirjoittaa* "write", *toimittaja* "journalist"), and chose ten topics: *hallitus, presidentti, venäjä, puolue, ulkopolitiikka, talouspolitiikka, turvallisuuspolitiikka, maahanmuuttopolitiikka, perussuomalainen, pääministeri*. We left out *suomi* (Finland) and *yhdysvalta* (United States), although they ranked high, because many tweets containing these words are not about politics. Instead, we chose *pääministeri* (prime minister) and *maahanmuuttopolitiikka* (immigration policy), which were also among the top 50 words. TF-IDF per sentiment also showed what characterises each group: negative leads had words like *arvostella* (criticise), *syyttää* (accuse) and *leikkauspolitiikka* (austerity policy), positive leads *hyvä* (good), *kehua* (praise) and *tyytyväinen* (satisfied). TF-IDF was repeated with a larger dataset but the

chosen keywords did pop up in the top 50 in the second run as well, so we did not change them.

Twitter data per keyword. For each chosen topic we collected a CSV of tweets with the word as the search term; immigration was searched with the broader *maahanmuutto*. For the parties we used one search term each (e.g. *SDP*, *kokoomus*, *vihreät*); the Centre Party was searched as *kepu*, because *keskusta* usually means "city centre".

Topic and party analysis (`analysis.py`, `analysis_parties.py`). For each topic, party and year we counted the leads or tweets of each sentiment and their shares. For the news, a topic matches if the word appears in the lemmas, so inflected forms and compounds are counted. For the parties, *keskusta* alone mostly means "city centre" and *vihreä* also means the colour green, so the Centre Party was matched only with *keskustapuolue*, *keskustalainen* and *kepu*, and the Greens with the exact lemma *vihreä*. As the main measure we used the **proportion of negative texts (%)**, which is easy to understand for non-technical readers.

Visualisation. `visualization_yearly.py` draws 100% stacked bar charts of the sentiment shares per year, with news and Twitter side by side, for each topic and party (Figure 1). `visualization_blog.py` draws the charts for the blog: the share of negative texts per year, dot charts comparing news and Twitter per topic and per party, and small multiples of each party over time with government periods and elections marked (Figure 2). `index.py` runs the analysis and visualisation steps in order.

5. Results

The news stayed calm but Twitter became more negative over 10 years. 88% of the news leads were neutral and only 8% negative. The yearly share of negative leads stayed between 5% and 10%. On Twitter, 41% of the topic tweets were negative, and the share rose from 28% in 2013 to 49% in 2022 (Figure 1). The growth came from both other groups: positive tweets fell from 18% to 11% and neutral tweets from 54% to 41%.

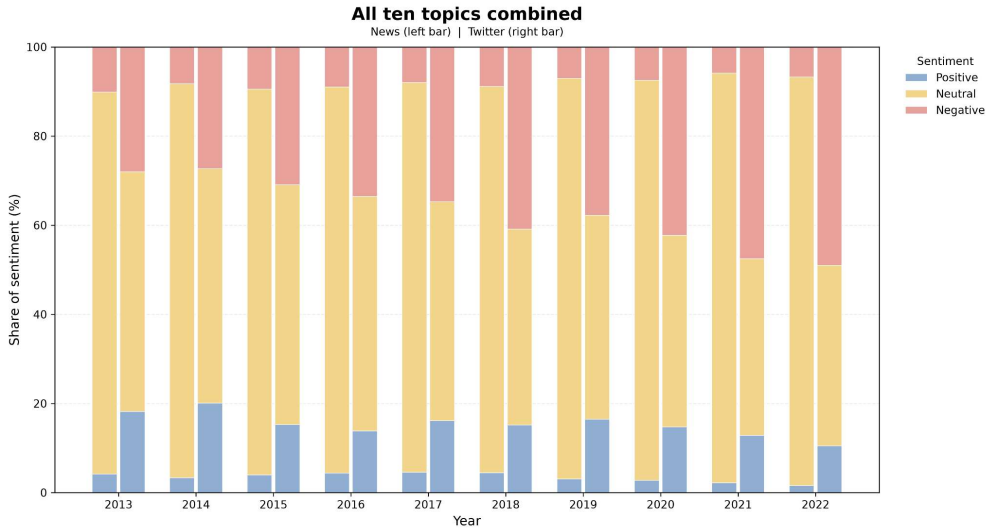


Figure 1. Sentiment shares per year for all ten topics combined: news (left bar) vs Twitter (right bar).

Topics. In the news, every topic was 3–10% negative. On Twitter the differences were large: the Finns Party (48% negative), the government (46%), immigration (44%) and Russia (40%) were the most negative topics, while security policy (14%) and the president (24%) were the calmest and the only topics with more positive than negative tweets. In 2022, tweets about Russia rose to 5,748, about six times the usual number, and 51% were negative, while only 7% of the news leads about Russia were negative.

Parties. In the news, 4–11% of the leads about each party were negative. On Twitter, negative tweets grew most for the parties in government (Figure 2). All five parties of the 2019–2023 government (SDP, Keskusta, Greens, Left Alliance, RKP) reached 42–66% negative tweets in 2022, compared with 36% for Kokoomus and 28% for the Finns Party, who were in opposition. Keskusta rose from 24% in 2014 to 66% in 2022. Liike Nyt had the lowest share (17%), but with few tweets.

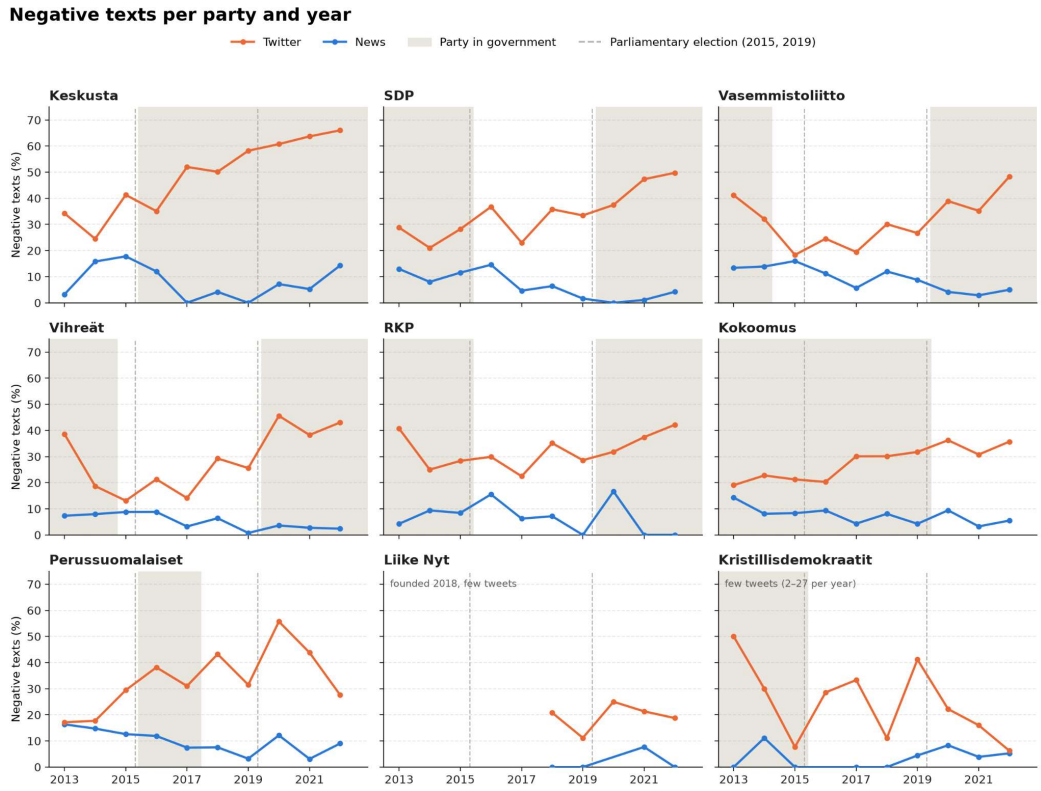


Figure 2. Share of negative texts per party and year. Grey: party in government; dashed lines: parliamentary elections.

6. Limitations

- **Model domain and validation.** FinBERT-FinnSentiment was trained on social media text, not news, so it may read formal news language as more neutral than a human would.

- **Threshold.** The 0.98 rule affects the absolute shares. The comparison between the sources is more reliable than the absolute numbers, because both use the same rule.
- **Different selection.** News leads were matched by lemmas and tweets by search terms. Tweets are not lemmatized, so a search for the base form misses inflected forms; for example, *perussuomalainen* found only 351 tweets.
- **Ambiguous names and small samples.** Keskusta has only 220 news leads after the stricter matching, and some groups have very few texts (e.g. the Christian Democrats 2–27 tweets per year, Liike Nyt 53 news leads), so their yearly changes are mostly noise.

7. Reflection on the canvas

What changed from the plan: Compared with our canvas, we extended the period to 2013–2022 to match the Twitter data, used leads instead of titles, and replaced the department filter with keyword filtering (section 3). Instead of training a model, we used a pretrained Finnish sentiment model, because adapting other BERT-based approaches to Finnish was not feasible within this project (section 4). We also extended the target group to the general audience (section 1).

What worked: The ready-made lemmas in the YLE corpus made keyword matching possible in Finnish language, and TF-IDF gave us better keywords than word frequencies. Using the same model and threshold for both sources made the comparison consistent. The sentiment model also produced good and explainable results, in general.

What did not work / what we would do differently: The Finnish language limited the models we could use, and the missing department tag forced us to change how we selected the news. We did not test the sentiment model on news specifically with fine tuning etc. Next time or with more time we would hand-label a sample of news leads to check the model and the threshold, and lemmatize the tweets so that both sources are matched the same way.

Future steps: Extend the data to 2023 to cover the parliamentary election, and compare other Finnish news and social media sources, including Swedish-language ones.

Learning outcomes: We learned how to preprocess a large annotated news corpus and tweet data, how to analyse and compare sentiment between two very different sources, and how to choose and present data so that it meets the needs of our target audience. We also learned how to apply a pretrained Finnish language model (FinBERT) to classify the sentiment of Finnish texts.

References

Laitinen, M., Lundberg, J., Levin, M. & Martins, R. (2018). The Nordic Tweet Stream. CEUR-WS.org/Vol-2084/short10.pdf · Yle Finnish News Archive, Kielipankki, kielipankki.fi/corpora · fergusq/finbert-finnsentiment, Hugging Face · Election results: Statistics Finland, stat.fi/fi/tilasto/evaa